GASTROENTEROLOGY · RNT MEDICAL COLLEGE, UDAIPUR · STUDY OPTIONS, NOVEMBER 2026

Which AI studies can our department realistically do?

We cannot follow patients for 5–10 years, and we have no digital records. That rules out long-term prognosis studies, but not AI research. This brief proposes short studies built around our busy hepatology ward, plus automation projects with published evidence behind them.

Follows the pilot summary on survival models. Every proposal below is based on published studies (references at the end). Patient numbers are placeholders until we have the ward's real admission figures.

Long follow-up is not needed

Studies can measure outcomes **during the admission**, at **one phone call at day 28 or 90**, or at **a single time point** (diagnosis).

One ward cohort, three or four papers

A single prospective register of cirrhosis admissions can answer questions on mortality, kidney injury and SBP together.

Start with the register

A free electronic case form with automatic score calculation and follow-up reminders solves the missing-records problem and supports every study here.

THE KEY POINT

Three study designs that fit a government hospital

Our earlier pilot looked at 5–10-year survival only because that old dataset was built that way. Most useful bedside questions in hepatology are short-term.

Design	Follow-up needed	Example question
Outcome during admission	None after discharge	Will this patient die, develop kidney injury or rebleed during this admission?
Short horizon	One phone call at day 28 or 90	Will this patient be alive at 28 days? The large Indian ACLF consortium uses 90-day follow-up.[1]
Diagnostic	None at all	Does this patient with ascites have SBP? Does this patient have high-risk varices?

Missing digital records are less of a barrier than they seem. In a prospective study we collect our own data from day one on an electronic form. REDCap is free for academic institutions; KoboToolbox and ODK work offline on a phone. Data collected this way are usually better than old hospital files.

HOW LONG WOULD IT TAKE?

The number that decides feasibility is events, not patients

To test whether a score or model works in our patients we need at least **100**, preferably **200**, patients who reach the outcome (for example, death).[2] Enter the ward's figures to estimate the duration.

Study duration estimator

Printed with example figures (40 admissions a month, 25% die within 28 days). The online version accepts the ward's real numbers.

Cirrhosis / ACLF admissions per month

40

Percentage who die within 28 days

25

Percentage we can enrol and follow

80

8.0
deaths per month in the study

13 months
to reach 100 deaths (minimum)

25 months
to reach 200 deaths (preferred)

Example figures only. Replace them with RNT's admission register counts.

CATEGORY 1

Short AI studies in the hepatology ward

All four use the same patients and the same case form. Study 1 is the core; studies 2 and 3 need only a few extra fields.

Recommended core study

1. Predicting short-term death in decompensated cirrhosis and ACLF

QUESTION

Do existing scores (AARC, CLIF-C ACLF, MELD-Na, MELD 3.0) correctly predict in-hospital and 28-day death in our patients? Does a machine-learning model add anything?

EVIDENCE IT MATTERS

The AARC score, built on 1,402 Indian ACLF patients, beat MELD and CLIF-SOFA (AUROC 0.80).[1] An AARC machine-learning model on 2,481 patients was 7% more accurate than AARC for 30-day death.[3] Both come from one consortium, and the standard AARC score already performs strongly,[4] so independent testing in a new region is the gap. Short-term deaths are common: in a Mumbai cohort, more than one-third of patients with a first alcohol-related decompensation died within 90 days.[7]

FOLLOW-UP

Admission plus one phone call at day 28 (optionally day 90).

ADDED VALUE

Also tests the ICU model we are developing on US intensive-care data (MIMIC-IV, eICU) in RNT patients.

Nested in study 1

2. Predicting acute kidney injury during admission

QUESTION

From day-1 data, which patients will develop AKI or HRS-AKI (ICA-AKI criteria) within 7 days?

EVIDENCE IT MATTERS

Machine-learning models on US ICU databases reached AUC 0.81–0.83 and beat MELD and SOFA.[5,6] All were retrospective, and the authors call for prospective external validation.

FOLLOW-UP

None after discharge.

Nested, diagnostic

3. Ruling out SBP before the ascitic fluid result

QUESTION

Can routine blood and clinical data identify patients with ascites who are very unlikely to have SBP?

EVIDENCE IT MATTERS

A model on 1,399 inpatients reached AUROC 0.82 with 93% sensitivity.[8] A 2024 US model showed a high negative predictive value and was aimed at resource-limited settings. About one-third of SBP is missed when paracentesis is not done.[9]

FOLLOW-UP

None. Our patients with ascites undergo paracentesis anyway, which gives the reference standard. The tool would help triage, for example in district hospitals, not replace paracentesis.

If FibroScan is available

4. Predicting high-risk varices without endoscopy

QUESTION

Do the Baveno VII rules (liver stiffness, spleen stiffness, platelets) and a machine-learning model correctly identify high-risk varices, using endoscopy as the reference?

EVIDENCE IT MATTERS

Spleen and liver stiffness models perform well, but mostly in hepatitis B and Chinese cohorts.[10,11] Validation in alcohol-predominant Indian patients is limited.

FOLLOW-UP

None. Mostly stable outpatients rather than ward patients.

CATEGORY 2

AI and automation for the department, by strength of evidence

Proven in randomised trials — Mostly purchases or service changes, not research projects

AI polyp detection during colonoscopy

A meta-analysis of 21 trials (18,232 patients) found adenoma detection rose from 35.9% to 44.0%.[12] It does not find more *advanced* lesions, it increases removal of harmless polyps, and real-world results are mixed.[13] A commercial system is worthwhile only with high colonoscopy volume.

AI reading of capsule endoscopy

Validated on 6,970 patients from 77 centres. A 2024 trial cut reading time from 33.7 to 3.8 minutes per study.[14] Relevant only if we perform capsule studies.

Bowel-preparation instructions by phone app

A meta-analysis of 8 trials (mostly WeChat, China) found better bowel preparation and lesion detection.[15] **Study opportunity:** a WhatsApp version in Hindi at RNT is cheap, quick, and publishable as a randomised trial. We found no Indian trial.

Validated tools, low cost

EncephalApp Stroop for covert hepatic encephalopathy

A smartphone test with AUROC 0.88 in 167 patients; a shorter QuickStroop version also exists.[16] **Study opportunity:** the paper-based test (PHES) depends on literacy, so validating the app in rural, Hindi-speaking patients would be a real contribution.

Automatic quality metrics from endoscopy reports

Software that reads free-text colonoscopy and pathology reports extracts adenoma detection rate with about 98% sensitivity and 99.7% specificity.[17] The first step for us is a structured digital endoscopy report template.

Infrastructure for everything above

Electronic cirrhosis admission register with automated follow-up

MELD 3.0, AARC and CLIF-C calculated automatically at entry; WhatsApp or SMS reminders and a daily list of patients due for day-28 and day-90 calls. A Cochrane review found that text-message reminders improve attendance.[18] This directly addresses poor follow-up of rural patients.

Not recommended yet

Building our own image-AI models (needs large labelled image sets and engineering support), and using large language models such as ChatGPT for clinical decisions (evidence still thin).

SUGGESTED PLAN

Three steps

Months 0–1

Set up the register

Electronic case form built from our existing variable list; ethics committee submission; train residents in data entry.

Months 1–18

Run the ward cohort

Study 1 (short-term death) with studies 2 (kidney injury) and 3 (SBP) nested in it. Day-28 phone follow-up.

In parallel

Quick projects

WhatsApp bowel-preparation trial, and EncephalApp validation in Hindi-speaking patients.

> **Decisions and figures needed from the department**
>
> - Monthly admissions of decompensated cirrhosis and ACLF, and a rough in-hospital death rate.
> - Whether the department has FibroScan (for study 4).
> - Monthly colonoscopy and capsule endoscopy volumes.
> - Whether RNT already has REDCap, and who can lead data collection on the ward.

REFERENCES

Sources

Figures were checked against abstracts and summaries. Read the full papers before citing them in a protocol.

1. Choudhury A, et al. AARC-ACLF score derivation and validation (1,402 patients, 90-day follow-up). *Hepatol Int* 2017. Record
2. Collins GS, Ogundimu EO, Altman DG. Sample size considerations for the external validation of a multivariable prognostic model: a resampling study. *Stat Med* 2016;35:214–226 (at least 100, preferably 200 events).
3. Verma N, et al. Artificial intelligence-based outcome prediction in ACLF (AARC). *Liver Int* 2023;43:442–451. Record
4. CAP-ACLF: comparative accuracy of prognostic models for short-term mortality in ACLF. Record
5. XGBoost prediction of AKI in cirrhosis, MIMIC-III/IV (950 patients, AUC 0.833). *Int Urol Nephrol* 2023. PDF
6. Early AKI prediction in critically ill cirrhosis, MIMIC-IV (1,240 patients, AUROC 0.808). arXiv 2025. arXiv
7. Outcome of individuals with alcoholic cirrhosis hospitalized with first decompensation and their predictors. *Indian J Med Res*. Article
8. Hu C, et al. Explainable machine learning model for predicting SBP in cirrhotic patients with ascites. *Sci Rep* 2021;11:21639. PMC
9. Decision tools for diagnosing SBP: systematic review and meta-analysis. Article
10. Liver and spleen stiffness model for high-risk varices. *World J Gastroenterol* 2023;29:4072. Abstract
11. Multicentre machine-learning model for high-risk varices in HBV cirrhosis. *Front Med* 2026. Article
12. Hassan C, et al. Computer-aided detection in colonoscopy: meta-analysis of 21 randomised trials. *Ann Intern Med* 2023. Summary
13. Patel, et al. Real-world (non-randomised) studies of computer-aided detection. *Endoscopy* 2024 (abstract). Abstract

14. Ding Z, et al. Gastroenterologist-level identification of small-bowel diseases by capsule endoscopy using deep learning. *Gastroenterology* 2019;157:1044–1054; and AI-assisted capsule reading in suspected small-bowel bleeding, *Lancet Digit Health* 2024. PubMed
15. Effect of WeChat-based education on bowel preparation: meta-analysis of randomised trials. PMC
16. Bajaj JS, et al. Validation of EncephalApp, a smartphone Stroop test, for covert hepatic encephalopathy. *Clin Gastroenterol Hepatol* 2015; QuickStroop 2023. Record
17. Natural language processing to measure colonoscopy quality metrics. *JMIR Med Inform* 2022. PDF
18. Gurol-Urganci I, et al. Mobile phone messaging reminders for attendance at healthcare appointments. *Cochrane Database Syst Rev* 2013;CD007458.

Dept of Gastroenterology, RNT Medical College, Udaipur. Planning document for discussion; not a protocol.

Pilot work and code: github.com/atultiwari/cirrhosis-ml-poc